SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID - SIH26143**
- **Problem Statement Title -** Leveraging satellite imagery to determine Oil spills at sea along with AIS data correlations to identify vessel responsible for the spill.
- **Theme -** Disaster Management
- **PS Category -** Software
- **Team ID -** FullSnack
- **Team Name -** FullSnack



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ORICEAN : AI Powered Oil Spill Intelligence

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PROPOSED SOLUTION

PLATFORM / ROLE	KEY FEATURES	WHAT IT ENABLES
Public User (Awareness & Reporting)	• View public spill detections & maps • Explore affected areas & timelines • Report new sightings or incidents • Access public dashboards & statistics	Transparency, awareness and community engagement
Analyst (Investigation)	• Create & manage investigations • Interactive map, AOI & time selection • View detections, trajectories & attribution	End-to-end investigation and decision support
Monitoring Officer (Monitoring & Alerts)	• Real-time spill alerts & notifications • Monitor ongoing incidents • Trend analysis & hotspot monitoring	Proactive monitoring and rapid response

AI-Powered Detection	Automated oil spill detection from Sentinel-1 SAR imagery.
Source Reconstruction	Wind & current-driven backtracking to identify probable spill origin.
AIS Correlation & Attribution	Correlate AIS vessel trajectories with source region and spill timeline.
Explainable Ranking	Multi-factor scoring with transparency and visual explanations.
Verification & Intelligence	Cross-verify with satellite re-detections and news/incident reports.
Dashboards & Insights	Interactive dashboards, timelines, heatmaps & downloadable reports.

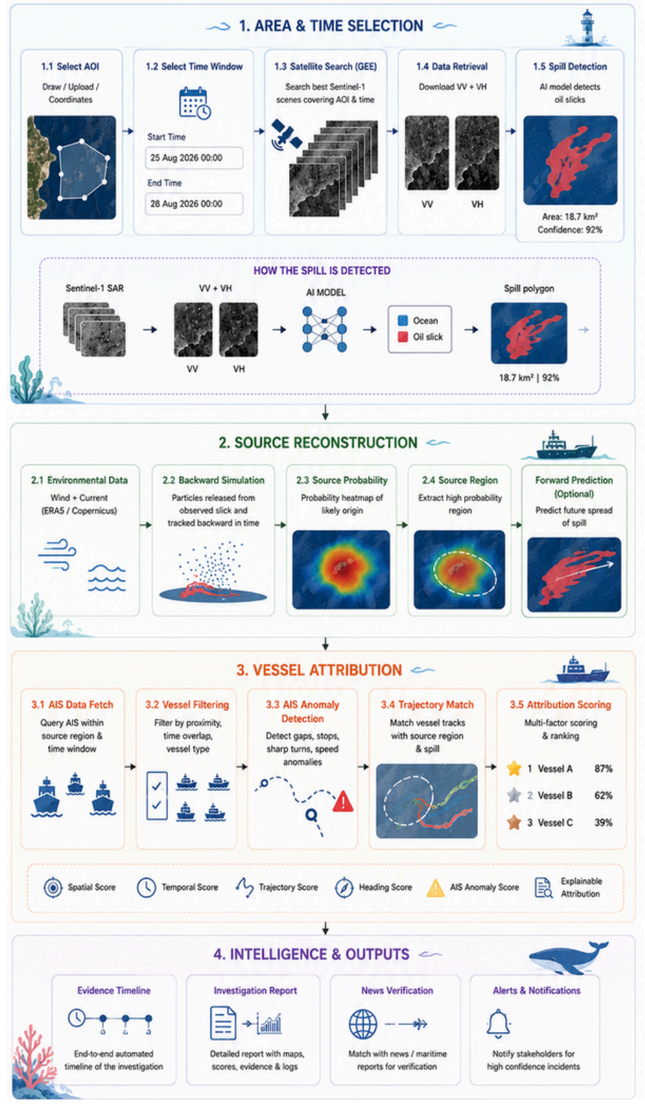
OUR UNIQUENESS & INNOVATION

AI + Satellite Fusion
First-time AI automation using Sentinel-1 SAR and deep learning for oil slick detection.

Physics-Informed Modelling
Wind & current-driven trajectory modelling to reconstruct the spill origin.

Forensic Attribution
AIS trajectory correlation with explainable scoring to identify potential polluting vessels.

Verified Intelligence
Integration of satellite re-detections and public reports to flag unreported incidents.



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TECHNICAL APPROACH

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TECHNOLOGY STACK

1. FRONTEND
React · TypeScript · Vite · Tailwind CSS

2. BACKEND
Python · FastAPI · SQLAlchemy · Redis

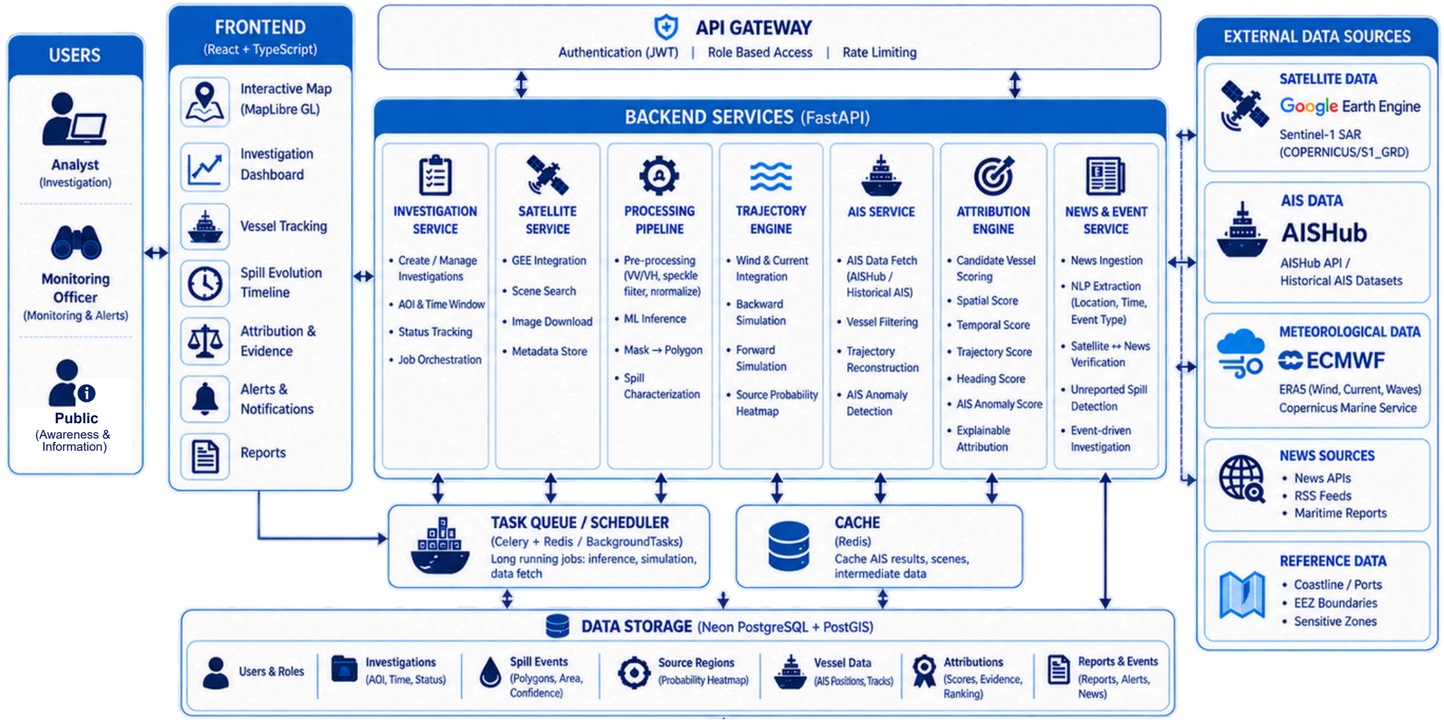
3. DATABASE
NEON · PostgreSQL · PostGIS

4. DATA PROCESSING / ML
PyTorch · NumPy · Pandas · Scikit-learn

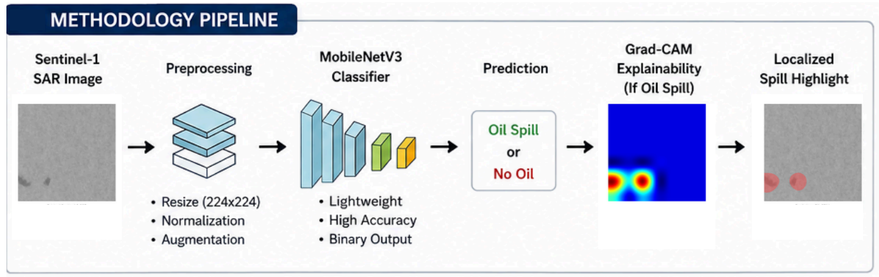
5. APIS & DATA SOURCES
AIS Data (AISHub API) · ECMWF API (Weather) · Google Earth Engine (Satellite Data) · OpenWeather API · Nominatim API (Geocoding)

6. GEOSERVICES & TOOLS
Google Earth Engine · Copernicus Marine Service · Natural Earth (Base Maps) · QGIS (Analysis)

SYSTEM ARCHITECTURE



METHODOLOGY PIPELINE



- High Accuracy: 97.47% on unseen test data
- Excellent Recall: 96.75% (Minimizes missed oil spills)
- Very High ROC-AUC: 0.9973 (Strong discriminative ability)
- Grad-CAM based explainability for transparent AI decisions
- Lightweight & Fast inference (MobileNetV3)
- Ready for deployment on large-area Sentinel-1 imagery

ML IMPLEMENTATION

Dataset
- Sentinel-1 SAR Images
- Oil Spill (Class 1)
- No Oil (Class 0)
- Balanced Dataset

Model
- MobileNetV3 (Large)
- Pretrained on ImageNet
- Modified for Binary Classification
- Input Size: 224 x 224

Training Setup
- Optimizer: AdamW
- Loss Function: CrossEntropyLoss
- Scheduler: ReduceLROnPlateau
- Epochs: 30
- Batch Size: 16

loss:	0.08927730438723952
accuracy:	0.9747292418772563
precision:	0.9571045576407506
recall:	0.967479674796748
f1:	0.9622641509433962
roc_auc:	0.9972899728997289

Classification Report

Class	Precision	Recall	F1-score	Support
No Oil	0.98	0.98	0.98	739
Oil Spill	0.96	0.97	0.96	369
Accuracy			0.97	1108
Macro Avg	0.97	0.97	0.97	1108
Weighted Avg	0.97	0.97	0.97	1108

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FEASIBILITY AND VIABILITY

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FEASIBILITY

- Builds on existing/government infrastructure rather than requiring new satellites or sensors.
- India already has mature geospatial platforms such as **Google Earth Engine (GEE)** & **ISRO's Bhuvan**/NDEM and operational disaster-management workflows using satellite data and GIS.
- **INCOIS** already operates an Oil Spill Advisory Service, using trajectory models, ocean circulation, atmospheric data and GIS; validating the feasibility of environmental spill modelling.
- AIS/LRIT-based vessel monitoring infrastructure already supports maritime situational awareness, making vessel-track correlation technically practical.











Challenge	Mitigation
SAR look-alikes	Oil + no-oil + look-alike training/evaluation
Satellite revisit / Data Latency	Process latest available acquisition + multi-temporal analysis
Uncertain spill origin	Probability region instead of exact source
Large satellite scenes	AOI filtering + tiling + cloud/object storage
False vessel attribution	Multi-factor scoring + trajectory validation

AREA	EXISTING SOLUTIONS	KEY LIMITATION
Satellite Spill Detection	EMSA CleanSeaNet	Detection-focused; limited end-to-end attribution
Oil Drift Modelling	INCOIS OSA	Predicts movement, not the complete source workflow
AIS Vessel Tracking	VesselFinder, MarineTraffic	Tracks vessels but lacks spill-specific correlation
Geospatial Platforms	ISRO Bhuvan, GIS dashboards	Visualization-focused, not investigation-centric
Incident Information	IOTPF databases, public reports	Limited integration with satellite evidence

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IMPACT AND BENEFITS

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ALIGNED WITH GOVERNMENT SCHEMES & NATIONAL INITIATIVES

MARITIME AMRIT KAAL VISION 2047
- Strengthening India as a global maritime power
- Focus on technology, digitalisation & sustainability
- Enhancing maritime domain awareness

Oricean supports the vision of a safe, secure and sustainable maritime future.

MARITIME DEVELOPMENT FUND (₹25,000 CRORE)
- Dedicated fund to promote growth, innovation and competitiveness
- Supports projects that strengthen the maritime ecosystem

Oricean is a technology enabler for future-ready maritime infrastructure.

SAMUDRA MANTHAN – OFFSHORE EXPLORATION SCHEME (₹84,084 CRORE)
- Expanding India's offshore exploration & resource potential
- Requires robust environmental monitoring and risk management

Oricean enhances offshore safety through real-time spill intelligence.

DEEP OCEAN MISSION
- Building capabilities for ocean exploration & technology
- Promoting sustainable use of ocean resources
- Advancing the Blue Economy

Oricean contributes to ocean knowledge, safety and sustainability.

NATIONAL OIL SPILL DISASTER CONTINGENCY PLAN (NOSDCP)
- Framework for preparedness and response to oil spills
- Coordinated by Indian Coast Guard
- Multi-agency responsibility & response

Oricean strengthens the intelligence layer for faster and more informed response.

SAGARMALA PROGRAM
- Port-led development & maritime connectivity
- Modernisation of ports and logistics
- Sustainable & resilient coastal ecosystem

Oricean supports safer operations and environmental protection across ports.

FASTER DETECTION
Detects oil spills within minutes of satellite overpass, enabling rapid response.

TIMELY RESPONSE
Early alerts help agencies act faster and reduce environmental impact.

EFFICIENCY & UTILIZATION
Optimized deployment of patrol vessels and clean-up resources.

RISK REDUCTION
Identifies high-risk vessels and routes, reducing incident ilkelihood.

COLLABORATIVE OPS
Enhances coordination between agencies with shared intelligence.

DATA-DRIVEN DECISIONS
Evidence-backed insights for investigations, enforcement and policy making.

COMPLIANCE & ENFORCEMENT
Supports regulatory compliance and strengthens enforcement actions.

STRATEGIC ADVANTAGE
Strengthens maritime domain awareness and national preparedness.

ENVIRONMENT PROTECTION
Protects marine life, fisheries and sensitive coastal habitats.

COST SAVINGS
Reduces clean-up, recovery and compensation costs through early action.

SUPPORTS OFFSHORE GROWTH
Enables safe and sustainable offshore operations and exploration.

INDUSTRY PARTNERSHIP
Facilitates collaboration with ports, oil companies and shipping stakeholders.

PUBLIC SAFETY
Protects coastal communities, livelihoods and public health.

GLOBAL CONTRIBUTION
Supports India's leadership in marine sustainability and global commitments.

THE OPPORTUNITY & IMPACT IN NUMBERS

7,517 km coastline

India's coastal ecosystems are vulnerable to marine pollution, and INCOIS explicitly identifies the importance of oil-spill trajectory prediction and **response along the Indian coast.**

Satellite monitoring is already operational globally

EMSA's CleanSeaNet demonstrates that SAR-based satellite monitoring can support oil detection and potential polluter identification, with alerts generally available within **20 minutes of satellite** overpass.

India is scaling maritime and offshore activity

Recent government initiatives include a **₹25,000 crore Maritime** Development Fund and **the ₹84,084 crore Samudra Manthan** offshore exploration scheme.

The intelligence gap remains

Existing systems focus on prediction, response and monitoring, but the end-to-end workllow from detection to source attribution and suspect vessel identification is still missing.

BENEFICIARIES

GOVERNMENT AGENCIES
Indian Coast Guard, MoPSW, MoES, Pollution Control Boards, DG Shipping, State Maritime Boards

PORTS & HARBOR AUTHORITIES
Major ports, minor ports and harbor authorities for safer and cleaner operations.

COASTAL COMMUNITIES
Fisherfolk, coastal residents and local communities whose livelihoods and health depend on clean oceans.

OIL & GAS COMPANIES
Offshore operators and energy companies to monitor, prevent and respond to incidents.

SHIPPING & MARITIME INDUSTRY
Shipping lines, vessel operators and logistics companies for risk reduction and compliance.

RESEARCH & ENVIRONMENT ORGANIZATIONS
Institutions working on ocean science, conservation and marine sustainability.

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RESEARCH AND REFERENCES

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1. Problem & Domain Research

- SIH 2026 – PS SIH26143 (NTRO): Defined the core workflow of satellite-based oil-spill detection, trajectory reconstruction and AIS-based vessel attribution. (SIH26143 Problem Statement)
- Focused our solution on: Automated Detection → Source Reconstruction → Suspect Vessel Ranking.
- Central Sector Scheme for Oil Spill Response (OSR) Facility: Run by the Ministry of Ports, Shipping and Waterways, this scheme provides financial support to major and non-major ports. It helps them buy Tier-I oil spill response equipment to manage localized minor spills quickly.

2. Indian Government Ecosystem

- INCOIS – Online Oil Spill Advisory System (OOSA): Studied India's existing oil-spill trajectory prediction and WebGIS advisory capabilities. (INCOIS OOSA Platform)
- Positioned our solution as an AI-driven investigation and vessel-attribution layer complementing existing response systems. OOSA integrates trajectory modelling, ocean/atmospheric models and GIS for oil-spill advisories.

3. Global Benchmarking

- EMSA CleanSeaNet: Studied Europe's satellite-based maritime monitoring ecosystem for oil-spill detection and vessel monitoring.
- Identified opportunities around AI-based detection, explainable localization, AIS anomaly analysis and automated evidence workflows.

4. Data & Technology Research

- Sentinel-1 SAR: Selected for its all-weather, day-and-night Earth observation capability, making it suitable for maritime monitoring.
- ESA Sentinel-1 Mission
- Google Earth Engine (https://earthengine.google.com)
- AIS Data: Researched historical vessel trajectories for reconstructing vessel movement and filtering potential suspects.
- Environmental Data: Wind and oceanographic variables support backward and forward spill trajectory modelling.

5. AI & Implementation Research

- Reviewed SAR oil-spill detection and segmentation approaches, including challenges involving oil-spill look-alikes.
- Evaluated Python geospatial tools, Google Earth Engine and PostGIS for scalable satellite processing, spatial analysis and visualization.